

CarbonLedger

Data Quality Plan

Per-plant metrics and a three-cycle improvement target

CL-DQP-001 · VERSION 1.0 · DRAFT FOR REVIEW

9

plants measured

3

improvement cycles

4

data tiers

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Summary & Contents

Purpose of this document

Defines how data quality is measured for each of the nine plants and sets a measurable three-cycle target for shifting estimated data toward metered data.

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Document control

Revision history

Version	Date	Description	Author (role)
1.0	03/10/2026	Initial release for review	Sole Engineer

Approvals

Role	Signature	Date
Sponsor (Manufacturer, Head of Sustainability)		
Reporting Lead (Manufacturer)		
Sole Engineer (all project roles)		
Reviewer (Faculty)		

Related documents

- CL-SRS-001 Software Requirements Specification (v1.0)
- CL-DTT-001 Decision Table and Test Set (v1.0)
- CL-DSG-001 Design Pack (v1.0)
- CL-CMP-001 Configuration Management Plan (v1.0)
- CL-DQP-001 Data Quality Plan (v1.0) · this document
- CL-PPL-001 Project Plan (v1.0)

Notice. This document is controlled; printed copies are uncontrolled. Figures stated as project data (data quality tiers of ±2%, ±8% and ±25%, nine plants, three scopes) come from the case study. Numbers in worked examples and test fixtures are synthetic, are labelled as such, and are not emission factors published by any authority.

1. Purpose and scope

This plan defines how data quality is **measured per plant**, how it is **disclosed**, and how estimated data is **shifted toward metered data over three reporting cycles**, with a target that has a measurable trajectory. It covers all **nine plants** (P1 to P9) and all **three scopes**.

Data quality matters here for a concrete reason: the uncertainty of the reported total is driven almost entirely by how much of it rests on estimates. Using the project's tier percentages (metered 2%, invoiced 8%, estimated 25%), an estimated item carries 12.5 times the relative uncertainty of a metered item. Improving data is therefore the only way to make the reported figure genuinely more certain; the arithmetic cannot do it.

2 Data quality tiers

Tier	Relative uncertainty	Evidence that qualifies	Typical sources
Metered	±2%	A meter reading with reading date and meter id	Plant sub-meters, fuel flow meters
Invoiced	±8%	A supplier bill or invoice for the quantity	Electricity bills, fuel purchase records, logistics invoices
Estimated	±25%	Any value derived from an assumption, a supplier declaration, or gap-filling	Distance assumptions, supplier declarations, carried-forward months
Missing	none (no value)	Nothing received	Resolved before close by receipt or gap-fill (becomes estimated, flag GAP_FILLED)

Rules: the tier follows the evidence (SRS FR-IN-03); a supplier declaration never upgrades a tier (FR-IN-06); a gap-filled item is always estimated (FR-CALC-06).

What each shift is worth. Moving an item one step reduces its relative uncertainty by:

Shift	Reduction (percentage points)	Share of the full estimated-to-metered gain
Estimated to invoiced	$25 - 8 = 17$	$17 / 23 = 74\%$
Invoiced to metered	$8 - 2 = 6$	$6 / 23 = 26\%$
Estimated to metered	$25 - 2 = 23$	100%

This drives the strategy in section 6: about three quarters of the benefit comes from simply obtaining the invoice, which needs no capital; metering is then added where an item is large.

3. Metrics per plant

All metrics are computed **from the issued report's lines**, so they are themselves reproducible. Notation: for plant p and cycle c , $E(i)$ is the emission of line i , $\text{tier}(i)$ its tier, $U(i)$ its uncertainty. Adjustment lines are excluded unless stated.

ID	Metric	Definition	Direction	Owner
DQ-1	Estimated share (primary KPI)	$S_{\text{est}} = \text{sum of } E(i) \text{ with tier estimated (including gap-filled and declared)} / \text{sum of all } E(i)$	Lower is better	Plant Manager
DQ-2	Metered share	$S_{\text{met}} = \text{sum of } E(i) \text{ with tier metered} / \text{sum of all } E(i)$	Higher	Plant Manager
DQ-3	Invoiced share	$S_{\text{inv}} = \text{sum of } E(i) \text{ with tier invoiced} / \text{sum of all } E(i)$	Information	Plant Manager
DQ-4	Shift to metered (M-SHIFT)	$(S_{\text{met}}(c) - S_{\text{met}}(0)) / (S_{\text{est}}(0) - S_{\text{est}}(c))$: the part of the reduction in estimated share that reappears as metered share	Higher	Plant Manager
DQ-5	Relative uncertainty	$R_{\text{rss}} = U_{\text{rss}} / E$ and $R_{\text{lin}} = U_{\text{lin}} / E$ for the plant (both disclosed)	Lower	Reporting Lead
DQ-6	Completeness at data cut-off	Expected items with a record stored by the cut-off / expected items	Higher	Plant Data Officer
DQ-7	Gap-filled items	Count and share of expected items resolved by gap-fill, by scope	Lower	Plant Manager
DQ-8	Late-data ratio	Sum of absolute adjustments attributed to the period / total E of that period (measured one cycle later)	Lower	Reporting Lead
DQ-9	Evidence coverage	Metered and invoiced lines with an evidence file / all metered and invoiced lines	Higher (must be 100% to close)	Plant Data Officer
DQ-10	Supplier declaration response rate	Declarations received by the cut-off / declarations requested (Scope 3)	Higher	Plant Manager and Procurement
DQ-11	Verify pass rate	Issued reports whose verify returns PASS / issued reports (system level, not per plant)	Must be 100%	Sole Engineer

DQ-1 to DQ-5 describe the *outcome* in the report; DQ-6 to DQ-10 are *leading indicators* that predict it; DQ-11 protects the integrity of all the others.

Computation. A database view joins **report_line** to **activity_record** for plant and tier:

```
CREATE VIEW v_dq_plant AS
SELECT rp.report_id, a.plant_id,
       SUM(l.emission_kg) FILTER (WHERE a.tier = 'ESTIMATED')
       / NULLIF(SUM(l.emission_kg), 0) AS s_est,
       SUM(l.emission_kg) FILTER (WHERE a.tier = 'METERED')
       / NULLIF(SUM(l.emission_kg), 0) AS s_met,
       SUM(l.emission_kg) FILTER (WHERE a.tier = 'INVOICED')
       / NULLIF(SUM(l.emission_kg), 0) AS s_inv
FROM report_line l
JOIN report rp ON rp.report_id = l.report_id
JOIN activity_record a ON a.record_id = l.record_id
WHERE NOT l.is_adjustment
GROUP BY rp.report_id, a.plant_id;
```

4. Per-plant scorecard

Each issued report is accompanied by a one-page scorecard with one row per plant, P1 to P9. The template is below; baseline values come from Cycle 0 and are entered by the system, not typed.

Plant	S_{est} (Cycle 0 = B)	Target C1	Target C2	Target C3	Actual	Status
P1	B(P1)	$0.60 \times B$	$0.30 \times B$	$0.10 \times B$	from report	RAG
P2	B(P2)	$0.60 \times B$	$0.30 \times B$	$0.10 \times B$	from report	RAG
P3 to P8	same rows, one per plant					
P9	B(P9)	$0.60 \times B$	$0.30 \times B$	$0.10 \times B$	from report	RAG
Portfolio	B (all plants combined)	$0.60 \times B$	$0.30 \times B$	$0.10 \times B$	from report	RAG

The scorecard also shows, per plant: DQ-2, DQ-4, DQ-5 (both values), DQ-6, DQ-7 by scope, DQ-9 and DQ-10, and the largest three uncertainty contributors (section 6.2).

Status rule (RAG). Green: actual meets the target. Amber: target missed but better than the previous cycle. Red: no improvement, or worse. Two consecutive reds trigger escalation to the Sponsor.

5 Three-cycle improvement target

5.1 Definition of cycles

Cycle 0 is the first reporting cycle (the report due on the first CRD). It sets the **baseline** for every plant. **Cycles 1, 2 and 3** are the next three reporting cycles. Baselines are measured, not assumed; no baseline number is invented in this plan.

5.2 Gap-closure rule

For every metric whose ideal value is known, the target is the baseline plus a fixed fraction of the remaining gap to the ideal:

 $\text{Target}(c) = B + g(c) \times (\text{Ideal} - B)$, with $g(1) = 0.40$, $g(2) = 0.70$, $g(3) = 0.90$

If a plant's baseline already meets the target, the target is "hold the baseline".

Why these fractions: the first cycle removes the easy gaps (bills that exist but were not entered, entry errors, unsent requests), so the largest step comes first; later cycles need meters, supplier agreements or process change, which have lead times, so the steps shrink (40, 30, then 20 points of the baseline gap). The last 10% is left as a residual, because some Scope 3 items may never be measurable. The fractions are set as a starting point and may be revised once after Cycle 0, under change control, when real baselines are known.

5. Three-cycle improvement target

5.3 Targets

Metric	Ideal	Cycle 1	Cycle 2	Cycle 3
DQ-1 Estimated share per plant	0	$0.60 \times B$	$0.30 \times B$	$0.10 \times B$
DQ-4 Shift to metered	not applicable	tracked	tracked	at least 0.50
DQ-5 Relative uncertainty per plant	not applicable	no increase	lower than C1	lower than C2
DQ-6 Completeness at cut-off	100%	$B + 0.40 \times (100\% - B)$	$B + 0.70 \times (100\% - B)$	$B + 0.90 \times (100\% - B)$
DQ-7 Gap-filled items, Scope 1 and 2	0	gap rule	gap rule	0 (the data exists at the plant)
DQ-8 Late-data ratio	0	$B \times 0.60$	$B \times 0.30$	$B \times 0.10$
DQ-9 Evidence coverage	100%	100%	100%	100%
DQ-10 Declaration response rate	100%	$B + 0.40 \times (100\% - B)$	$B + 0.70 \times (100\% - B)$	$B + 0.90 \times (100\% - B)$
DQ-11 Verify pass rate	100%	100%	100%	100%

DQ-9 and DQ-11 are not trajectories: they are gates and hold at 100% every cycle.

5.4 Worked trajectory (synthetic illustration, not plant data)

Take one plant with baseline shares of emissions: metered 10%, invoiced 40%, estimated 50%, so $B = 50\%$. The targets give 30%, 15% and 5% for estimated. A feasible path, with the worst-case relative uncertainty $R_{lin} = 0.02 \times S_{met} + 0.08 \times S_{inv} + 0.25 \times S_{est}$:

Cycle	Metered	Invoiced	Estimated (target)	M-SHIFT	R_{lin}
0 (baseline)	10%	40%	50%	-	$0.02 \times 0.10 + 0.08 \times 0.40 + 0.25 \times 0.50 = 15.9\%$
1	15%	55%	30%	$5 / 20 = 0.25$	$0.003 + 0.044 + 0.075 = 12.2\%$
2	25%	60%	15%	$15 / 35 = 0.43$	$0.005 + 0.048 + 0.0375 = 9.05\%$
3	40%	55%	5%	$30 / 45 = 0.67$	$0.008 + 0.044 + 0.0125 = 6.45\%$

Cycle 3 meets both targets (estimated 5% = $0.10 \times 50\%$; M-SHIFT $0.67 \geq 0.50$). The worst-case relative uncertainty falls from 15.9% to 6.45%, a reduction of 59%. This is the measurable trajectory the report discloses each cycle.

5.5 What changes in each cycle

Cycle	Main lever	Concrete actions
1	Estimated to invoiced (about 74% of the gain per item)	Chase every missing bill and invoice; enter backlog; replace assumed logistics distances with invoice data; fix entry errors found by DQ-6.
2	Estimated and invoiced to metered for the largest contributors; supplier engagement	Metering plan for the top uncertainty contributors (section 6.2); sub-meter or read existing meters; agree declaration format with suppliers.

5. Three-cycle improvement target

Cycle	Main lever	Concrete actions
3	Metered share and residual cleanup	Complete metering where justified; remove remaining gap-fill for Scope 1 and 2; review Scope 3 items that remain estimated and document why.

6 How to decide where to act

6.1 Responsibility

Plant Data Officers keep records complete and evidenced. Plant Managers own DQ-1 to DQ-4, DQ-7 and DQ-10 for their plant. The Reporting Lead owns DQ-5 and DQ-8 and the portfolio view. The Sole Engineer owns DQ-11 and the metric views.

6.2 Priority by contribution to uncertainty

Because totals add by root-sum-square, a few large uncertain items dominate. For each plant, rank lines by

 $P(i) = U(i)^2 / \text{sum of } U(j)^2$ (share of the plant's variance)

and work on the items at the top until they cover at least 80% of the plant's variance (a common Pareto convention). A large estimated item is attacked first; a small metered item never.

6.3 Choosing the lever

Situation	Action	Reason
Bill or invoice exists or can be requested	Convert to invoiced	No cost; 17 of the 23 points.
Item is large and recurring and a meter can be fitted or read	Convert to metered	Extra 6 points; worth the cost only for items high in $P(i)$.
Supplier data (Scope 3)	Ask for declaration; request the underlying invoice quantities	A declaration alone stays estimated; invoice quantities can reach invoiced.
No data can be obtained	Gap-fill, disclose, keep on the watchlist	Honest estimate with its uncertainty.

7. Supplier non-response procedure

Supplier non-response is a known, expected problem (risk R-01 in CL-PPL-001). The data quality response is a fixed routine tied to the cycle calendar (T = contract reporting date, in working days):

Day	Step	Owner
T-25	Request issued to every supplier whose declaration or invoice data is expected.	Plant Manager and Procurement
T-20	First reminder to non-responders.	Plant Data Officer
T-15	Second reminder, copied to the supplier's account contact; escalation to Procurement.	Plant Manager
T-14 (cut-off)	Non-responders recorded; item stays an Open Item.	System
Close	Open Item is gap-filled, tier estimated, flag GAP_FILLED; listed in the report with its uncertainty.	System
After close	Item stays on the Late-Data Watchlist; a declaration arriving later is processed under rule R10 (late adjustment).	Reporting Lead

A supplier that does not respond in two consecutive cycles is flagged; Procurement raises it as a commercial matter and the Sponsor considers asking the customer to accept an agreed fallback. DQ-10 is reported per supplier and per plant.

8

Quality gates

Gate	Check	Effect
At entry	Unit from list; quantity numeric and non-negative; date inside plant operating period; evidence file present for metered and invoiced; tier consistent with evidence type; duplicate detection (same plant, type, interval, quantity).	Record rejected with a message, or accepted with a warning.
Plausibility	Compare with the same item in the previous month and year; deviation threshold is configurable and set from the Cycle 0 observed variation , not guessed now.	Warning shown to the Plant Manager; not a hard stop.
Before close	No unresolved Open Item; evidence coverage 100% (DQ-9); Plant Manager confirmation for each plant.	Close refused until met (SRS FR-PC-02).
After report	Self-verify returns PASS (DQ-11).	Report not issued on FAIL.

9. Disclosure to the customer

Each issued report contains a **Data Quality section** generated from the same canonical data:

1. Totals by scope and for the whole company as $E \pm U (x\%)$, with the worst-case bound.
2. Share of emissions by tier, per scope and in total.
3. The list of gap-filled and declared items, with plant and size.
4. The per-plant scorecard (section 4) with the three-cycle target and the actual position.
5. The fixed method statement: tier percentages 2%, 8%, 25%; root-sum-square; independence assumption; factor uncertainty not included.

This keeps the disclosure honest: the customer sees the figure, how much of it rests on estimates, and whether that share is falling as promised.

10 Review and governance

Event	Frequency	Participants (roles)	Output
Plant data review	Monthly	Plant Manager, Plant Data Officer	Open items, evidence gaps, DQ-6 and DQ-9 check.
Cycle data quality review	At each close	Reporting Lead, Sole Engineer	Scorecard, RAG status, DQ-8 for the previous period.
Sponsor review	Each cycle	Sponsor, Reporting Lead	Decisions on metering spend, escalations, target revision (once, after Cycle 0).
Improvement action register	Continuous	Owners by role	Each action has owner, plant, lever, due cycle, expected effect on S_{est} .

11 Assumptions and limits

- Baselines are unknown until Cycle 0; the targets are therefore relative to each plant's own baseline.
- Tier percentages are used as given and apply to activity data; the authority's own factor uncertainty is not included.
- Shares are measured by **emissions**, not by record count, because emissions determine uncertainty.
- Target fractions (0.40, 0.70, 0.90) are a planning choice, revisable once under change control after Cycle 0.